

Arash Jalil Khabbazi, MS

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Profile

I am a third-year PhD student in Mechanical Engineering at Purdue, working on reinforcement learning (RL) and predictive control for buildings, including model predictive control (MPC) and data-driven predictive control (DDPC). I am also interested in applying broader machine learning (ML) methods to AI-enabled problems across domains.

Education

Purdue University, USA PhD in Mechanical Engineering — GPA: 3.89/4.0 Specialization in Computational Science and Engineering (CSE)	08/2023 – 07/2027
University of British Columbia, Canada MS in Mechanical Engineering — GPA: 4.0/4.0 (94%)	05/2021 – 08/2023
University of Tabriz, Iran BS in Mechanical Engineering — GPA: 4.0/4.0 (19.12/20) — Rank: 1/155+	08/2016 – 07/2020

Experience

Applied ML Intern – Tesla, Inc. – Sparks, NV, USA • Built supervised ML models (XGBoost, DNNs, SVR, and ensembles) for defect detection and quality assessment. • Developed data pipelines in MySQL and engineered features to extract high-signal inputs for ML models. • Contributed to deployment of a thermal imaging system for non-destructive inspection in drive-unit manufacturing. • Collaborated on projects projected to reduce testing costs by over \$1M annually through improved quality assessment.	05/2025 – 08/2025
Graduate Research Assistant (PhD) – Purdue University – West Lafayette, IN, USA • Derived a multi-zone formulation for HVAC control savings overestimation from cross-zone heat transfer (<i>ACC</i> 2026). • Developed an RL controller for residential HVAC using the OCHRE, saving 17.5% energy while maintaining comfort. • Reviewed 100+ publications on MPC and RL for HVAC, resulting in a review paper ; GitHub . • Created real-time dashboards for energy/comfort and ML models for occupancy detection (98% accuracy). • Supported commissioning of a university building by upgrading a Niagara-based building automation system. • Designed data-enabled predictive control (DeePC) for HVAC, matching MPC without an explicit system model. • Assisted in developing active current-limiting control for appliances to prevent breaker overloads in electrified homes.	08/2023 – 08/2027
Graduate Research Assistant (MS) – University of British Columbia – BC, Canada • Researched mixing gaseous hydrogen into natural gas distribution pipelines (MS thesis). • Performed CFD simulations and developed ideal- and real-gas mixture models in C; GitHub . • Worked with FortisBC Energy Inc. research teams to assess industry impacts as part of UBC H₂Lab projects. • Produced one journal paper and three conference papers based on research findings, earning two awards.	05/2021 – 08/2023

Selected Publications — Full list on [Google Scholar](#)

- A.J. Khabbazi** and K.J. Kircher. Small HVAC control demonstrations in larger buildings often overestimate savings. *American Control Conference (ACC)*, 2026.
- A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze and K.J. Kircher. Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review. *Applied Energy* (2025).
- E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani and K.J. Kircher. Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study. *CISBAT*, 2025.
- A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou and J. Quinn. Mixing hydrogen into natural gas distribution pipeline system through Tee junctions. *International Journal of Hydrogen Energy* (2024).

Skills

- Expertise:** Predictive Control, Reinforcement Learning, Data Structures & Algorithms, Convex Optimization, Control Theory, Machine Learning, Deep Learning, Generative AI, Industrial IoT, Buildings & HVAC, Thermodynamics
- Graduate Coursework:** Artificial Intelligence, Reinforcement Learning & Control, Optimal Control & Estimation, Convex Optimization, Statistical Methods, Industrial Internet of Things, Distributed Energy Resources, Analysis of Thermal Systems, Advanced Thermodynamics
- Programming & Tools:** Python, MATLAB, C, Git, Linux, Docker
- ML & Optimization:** PyTorch, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, CVX, CVXPY, Gurobi
- Data Analysis & Visualization:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Grafana
- Simulation & Testbeds:** Gymnasium, OCHRE, BOPTEST
- Databases & Integration:** MySQL (SQL), InfluxDB, Tridium Niagara, Modbus, IO-Link, MTConnect

Selected Honors & Awards

Best PhD Forum Presentation Runner-up Award at ACM BuildSys (2025) — ASHRAE Graduate Student Grant-in-Aid Award (2025) — Best Paper Award at CSME (2023) — Best Presentation Award at CSME (2022) — UBC Graduate Scholarship (2022) — UBC Dean's Entrance Scholarship (2021)